Muhammad Muizz Zafar

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Data Engineer | ETL Pipelines | Data Warehousing | AWS & Cloud Data Platforms

Skills

Languages:	Python, SQL, GoLang, Java, C#, C++
Data Engineering:	Data Warehousing, Data Modeling, Apache Airflow, PySpark, Apache Spark, FME, Power Designer AWS (S3, Lambda, Glue Crawlers, Athena), MSSQL Server, Oracle, Docker, Git, Linux, Datalake

Education

University of Ottawa — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

Work experience

Data Engineer — May 2024 - Present
Government Of Canada — Ottawa, ON

- Designed logical and physical data models using **SAP PowerDesigner**, defining **schema design**, **data architecture**, and **information architecture** for a centralized enterprise **data warehouse** supporting 20M+ records across operational and **research databases**.
- Built and maintained scalable **ETL/ELT pipelines** using **Python**, **SQL**, **PySpark**, **Apache Spark**, and **Apache Airflow** to ingest, transform, and load structured and semi-structured data from **Oracle**, **SQL Server**, **NoSQL** sources, and cloud storage into **Snowflake** and **Databricks**.
- Developed **quality control procedures** and automated validation frameworks to reconcile source-to-target transformations, implement business rules, and reduce data discrepancies by 60%.
- Orchestrated **containerized** data workflows using **Docker** and **Kubernetes**, enabling portable, reproducible deployments across development, UAT, and production environments.
- Integrated data from multiple **cloud computing** platforms, including **AWS** (**S3**, **Lambda**, **Glue**, **Athena**), **Azure**, and **GCP**, leveraging **BigQuery**, **Snowflake**, and **Databricks** to support high-volume analytics and **machine learning** workloads.
- Optimized complex **SQL** queries, **Spark** jobs, and warehouse transformations, reducing end-to-end pipeline runtimes by 40% and improving data availability for downstream users.
- Collaborated with business and research stakeholders to reconcile business requirements with **information architecture** needs and design technical solutions for highly complex **system integration** initiatives.
- Built interactive **Power BI** and **Tableau** dashboards to deliver **data visualization** of pipeline performance, data quality metrics, and operational KPIs, enabling stakeholders to identify and resolve issues 2x faster.
- Leveraged **Azure DevOps** and **Git** to implement **CI/CD pipelines**, automate testing, and promote code and infrastructure changes across environments.
- Applied **cloud computing** and **data engineering** best practices to support large-scale **R&D** initiatives involving **AI**, advanced analytics, and autonomous data-driven experimentation.

Full Stack Developer Intern — September 2023 - April 2024
University of Ottawa — Ottawa, ON

- Redesigned the University of Ottawa's emergency alert website with **C#** and **Angular**, introducing a modern, interactive UI. Used by over **20,000** students daily.
- Optimized existing **C#** REST APIs by reducing **SQL** query execution time for event data lookup by **30 %** and lines of code by **40 %**
- Revamped the event registration and admin webpages by using **Angular** material UI components to improve UI layout and reducing redundant lines of code by almost **50 %**

Data Engineer Intern — May 2023 - August 2023
Trend Micro Inc. — Ottawa, ON

- Designed and implemented a serverless data pipeline using **AWS Lambda, S3, Glue Crawlers, and Athena**, leveraging **AWS SAM** and **CloudFormation** for infrastructure-as-code and automated deployment.
- Automated metadata updates by configuring **Glue Crawlers** to detect changes in S3 datasets, eliminating manual schema management and improving data accessibility.
- Configured **AWS CloudWatch** and **Splunk** monitoring to track pipeline performance, logs, and failures, enhancing observability and system reliability.

Cloud Developer Intern — September 2022 - December 2022
Trend Micro Inc. — Ottawa, ON

- Developed serverless data processing pipelines using **AWS Lambda** to analyze client files, images, and code stored in **Amazon S3**, performing vulnerability scans using Anchore APIs and Grype.
- Implemented concurrent processing and robust error handling in **GoLang** using goroutines within Lambda functions, improving reliability and reducing data corruption in high-throughput environments.
- Enabled scalable data querying workflows by integrating **AWS Glue Crawlers** to automatically update metadata and **Amazon Athena** to query data stored in S3, reducing manual data processing efforts.